Lessons From Newton’s Laws: Field and Force

The top half of the table, used fifteen times

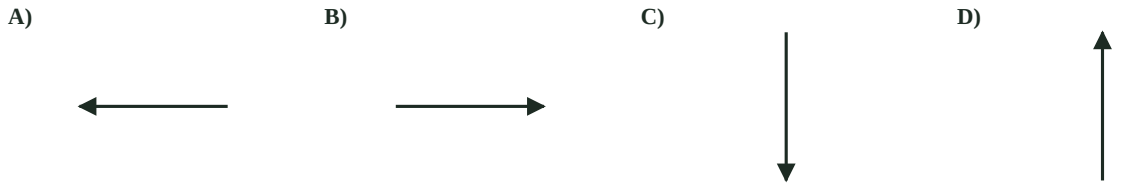
Everything here runs on the first three rows of the table in **Note 8**: $F = kq_1q_2/r^2$, $F_E = qE$, and $\mathbf{E} = kq_1/r^2 = F_E/q$. Keep that note open beside this one if you want the gravitational column to check yourself against.

1. In the diagram below, *P* is a point near a negatively charged sphere.



Figure 2. A point *P* beside a negatively charged sphere.

Which vector best represents the direction of the electric field at point *P*?



2. Which graph best represents the relationship between the magnitude of the electric field strength, *E*, around a point charge and the distance, *r*, from the point charge?

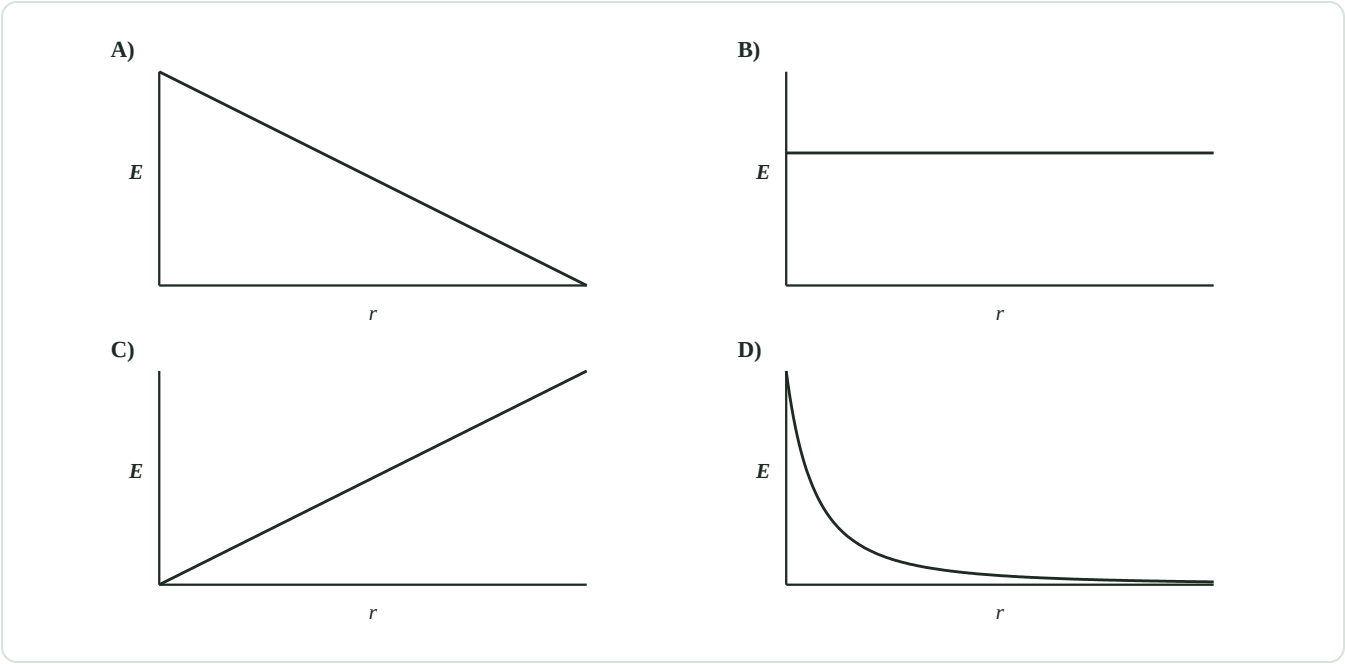


Figure 3. Four graphs of *E* against *r*.

3. In the diagram below, *A* is a point near a positively charged sphere.



Figure 4. A point *A* beside a positively charged sphere.

Which vector best represents the direction of the electric field at point *A*?

A) 

B)

C) 

D) 

4. Two positively charged masses are separated by a distance, *r*. Which statement best describes the gravitational and electrostatic forces between the two masses?

   A) Both forces are attractive.

   B) Both forces are repulsive.

   C) The gravitational force is repulsive and the electrostatic force is attractive.

   D) The gravitational force is attractive and the electrostatic force is repulsive.

5. The diagram below shows two identical metal spheres, *A* and *B*, separated by distance *d*. Each sphere has mass *m* and possesses charge *q*.

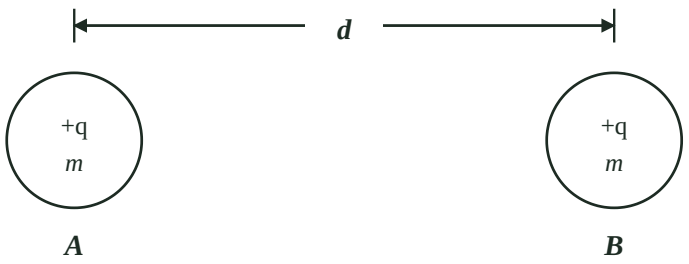


Figure 5. Two identical spheres, each of mass *m* and charge *q*.

Which diagram best represents the electrostatic force F_e and the gravitational force F_g acting on sphere *B* due to sphere *A*?

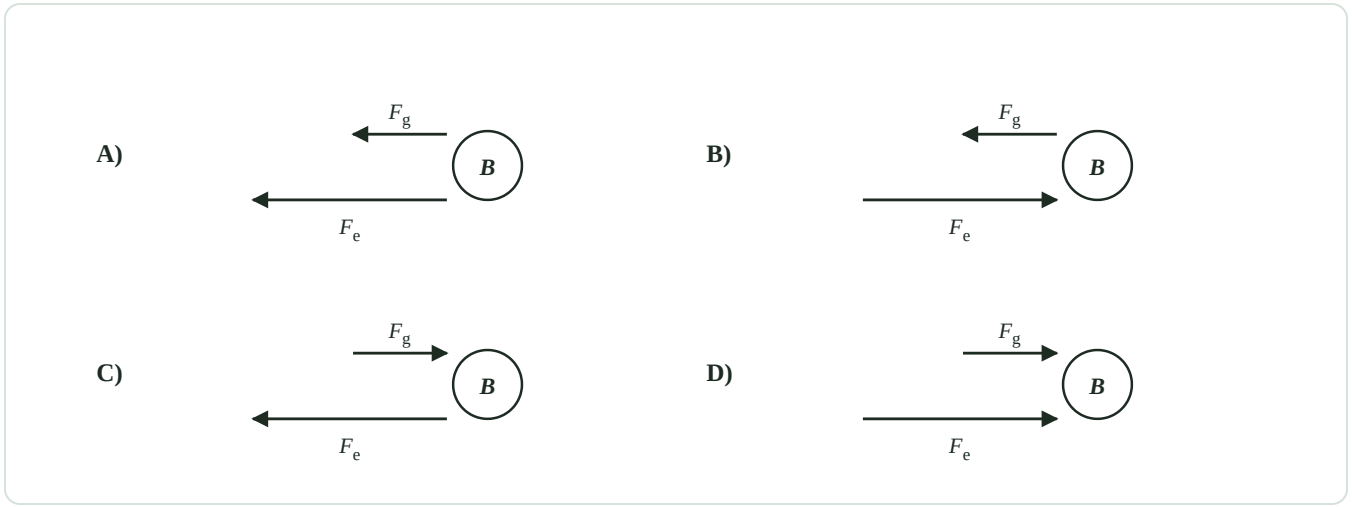


Figure 6. Four options. Sphere *A* is off to the **left** in every one of them, so an arrow pointing left is a force *toward A*.

6. Base your answer to the following question on the diagram below which represents two small, charged conducting spheres, identical in size, located 2.00 meters apart.

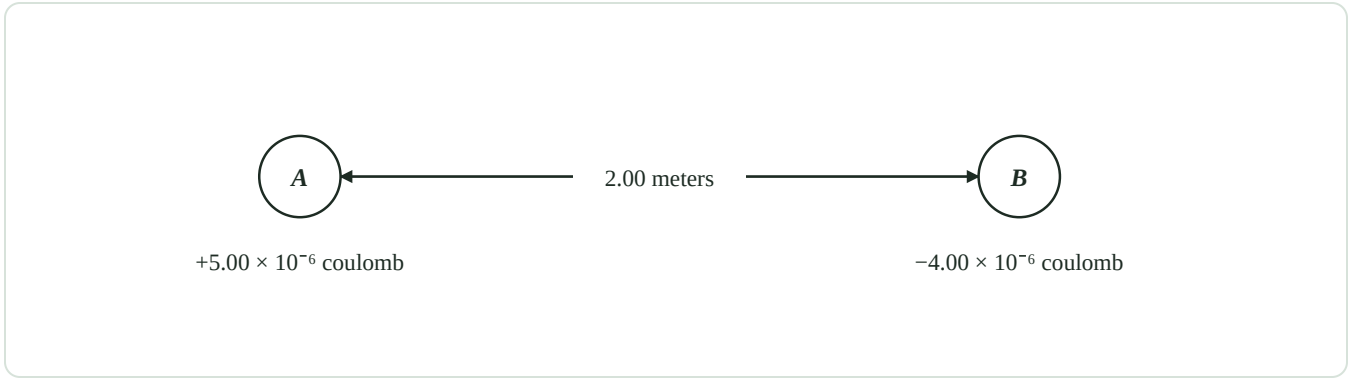


Figure 7. Two conducting spheres 2.00 m apart, one positive and one negative.

Which diagram best represents the electric field between the two spheres?

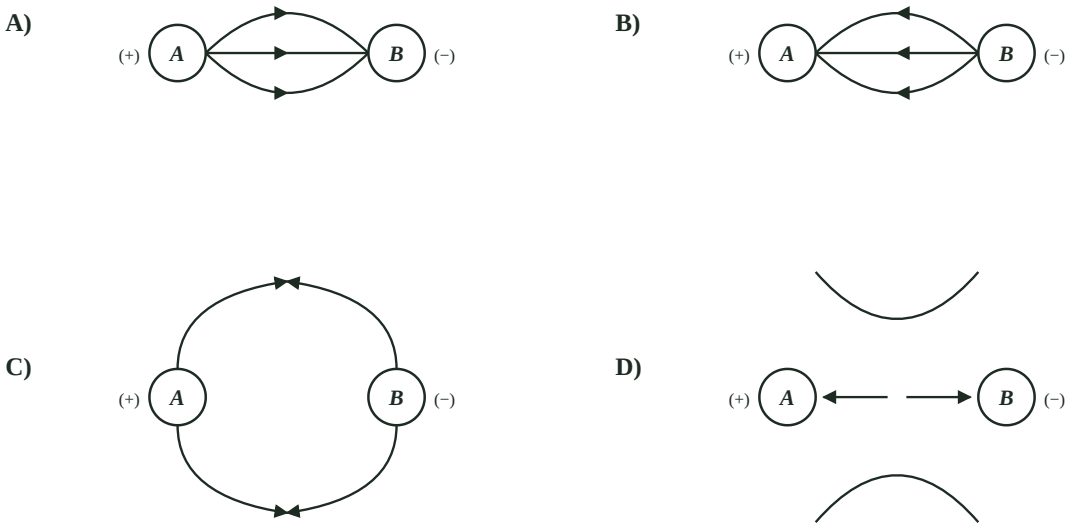


Figure 8. Four options. **Watch the arrowheads** — two of these four draw the same lines and differ only in which way round they run.

7. Base your answer to the following question on the diagram below which represents an electron being projected between two oppositely charged parallel plates.

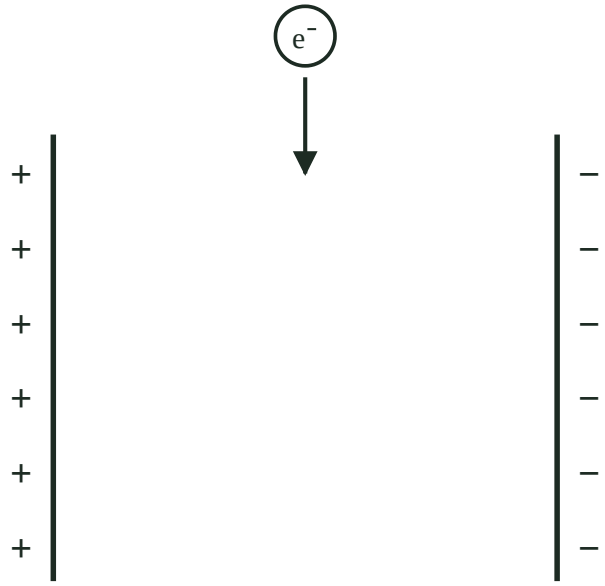


Figure 9. An electron entering the gap between two oppositely charged plates.

As the electron moves through the electric field, the magnitude of the electric force on the electron

A) decreases

B) increases

C) remains the same

8. The direction of the electric field at a point in space is defined as the direction of the force exerted by the field on a

A) test mass located at that point

B) magnetic north pole located at that point

C) negative test charge located at that point

D) positive test charge located at that point

9. Base your answer to the following question on the information and diagram below. Two small metallic spheres, *A* and *B*, are separated by a distance of 4.0×10^{-1} meter, as shown. The charge on each sphere is $+1.0 \times 10^{-6}$ coulomb. Point *P* is located near the spheres.

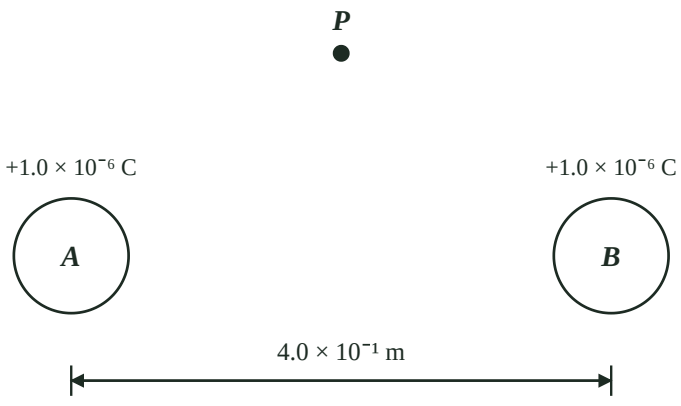
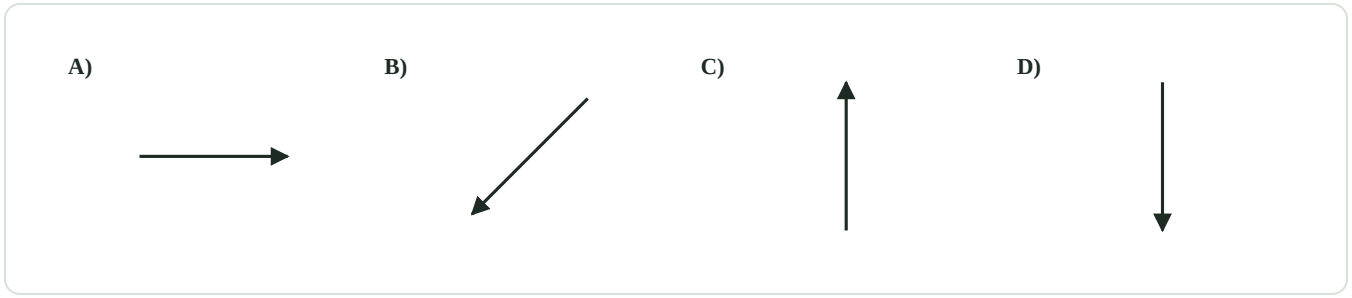


Figure 10. Two equal positive charges with *P* above them.

Which arrow best represents the direction of the resultant electric field at point *P* due to the charges on spheres *A* and *B*?



10. Two plastic rods, *A* and *B*, each possess a net negative charge of 1.0×10^{-3} coulomb. The rods and a positively charged sphere are positioned as shown below.

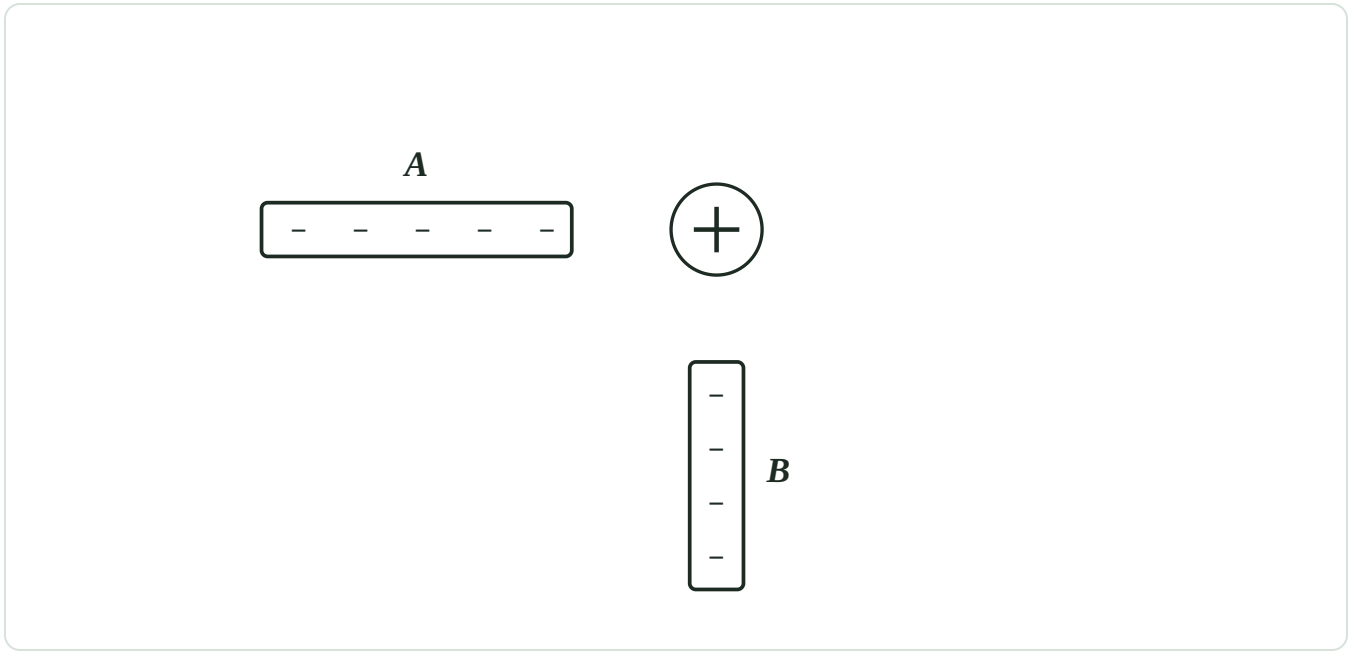
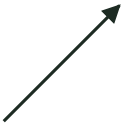


Figure 11. Two negatively charged rods and a positively charged sphere.

Which vector best represents the resultant electrostatic force on the sphere?

A)



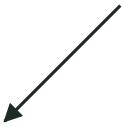
B)



C)



D)



11. The diagram below represents oppositely charged plates in an evacuated glass tube.



Figure 12. Two oppositely charged plates inside an evacuated tube.

Which diagram below represents the path of free electrons between the charged plates?

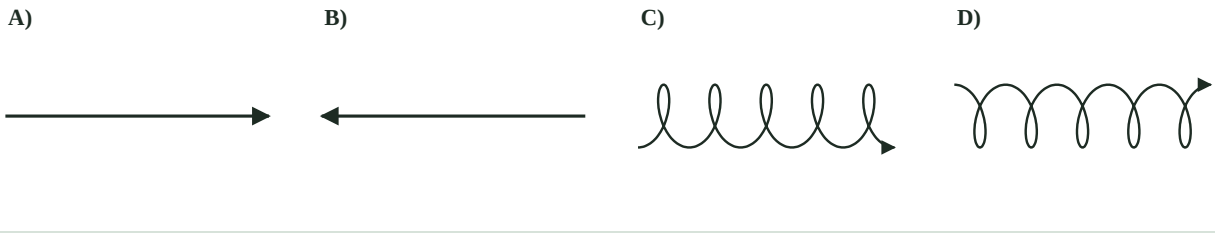


Figure 13. Four option paths.

12. In the diagram below, proton *p*, neutron *n*, and electron *e* are located as shown between two oppositely charged plates.

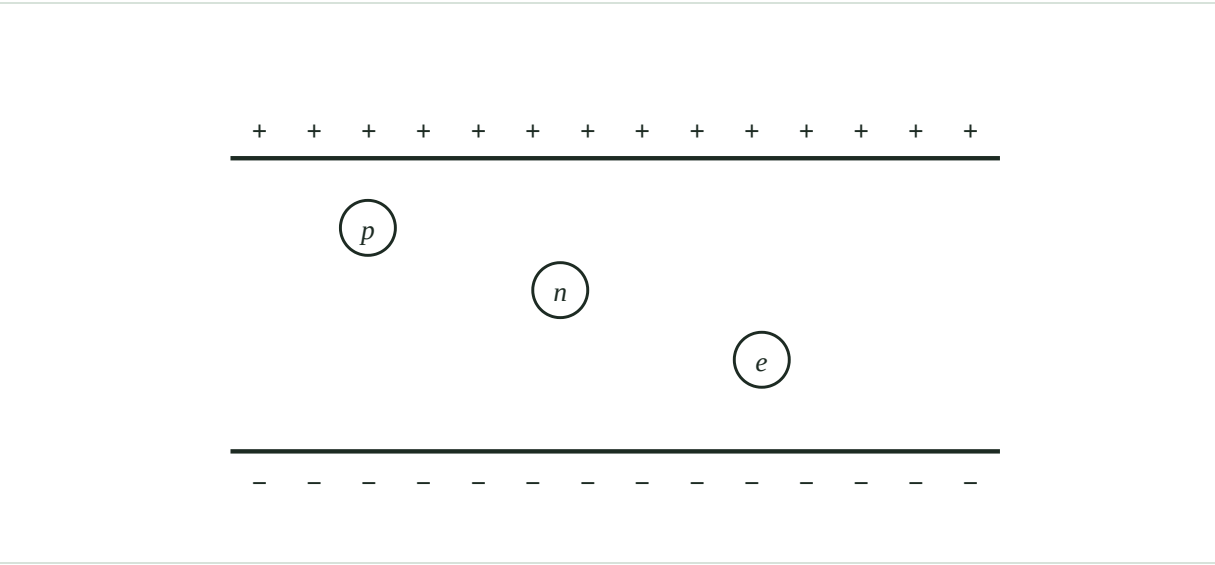


Figure 14. A proton, a neutron and an electron between charged plates.

The magnitude of acceleration will be greatest for the

A) neutron, because it has the greatest mass

B) neutron, because it is neutral

C) electron, because it has the smallest mass

D) proton, because it is farthest from the negative plate

The oil drop

Base your answers to questions **13** through **15** on the information and diagram below and on your knowledge of physics.

The diagram shows a negatively charged oil drop that is suspended motionless between two oppositely charged, parallel, horizontal metal plates. The electric field strength between the charged plates is 4.0×10^{4} newtons per coulomb. The 1.96×10^{-15}-kilogram oil drop is being acted upon by a gravitational force, F_g, and an electrical force, F_e.

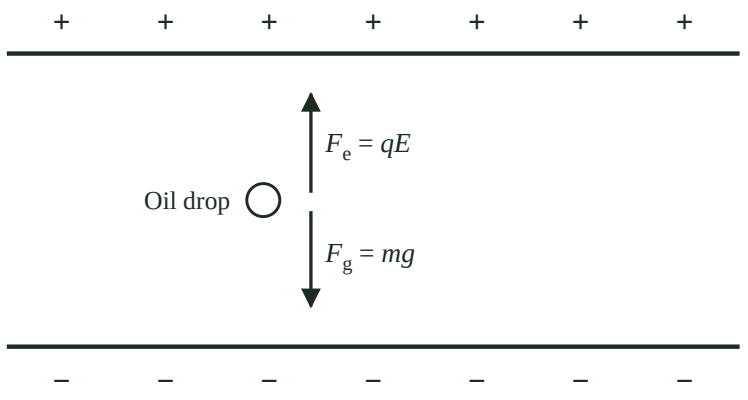


Figure 15. The drop hangs still, so the two forces on it are equal and opposite. **The two labels are his**, written on the diagram: $F_e = qE$ upward and $F_g = mg$ downward.

13. Calculate the magnitude of the gravitational force, F_g, acting on the oil drop. [Show all work, including the equation and substitution with units.]

14. Determine the magnitude of the upward electrical force, F_e, acting on the oil drop suspended motionless between the charged metal plates.

15. Calculate the net electric charge on the oil drop in coulombs. [Show all work, including the equation and substitution with units.]

Looking ahead

Note 10 is this same set of fifteen again, run on the bottom half of the table — potential and work instead of field and force. The questions will feel familiar and the arithmetic will be easier, because energy is a number and force is a vector.